

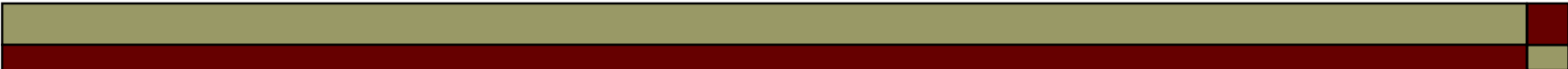
Software Construction and Development SE-312

LECTURE # 05

Individual software process (PSP)

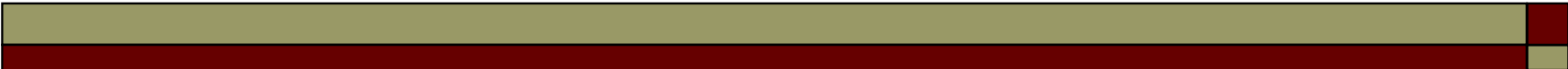
Team software process (TSP)

Course Instructor: Engr. Ruqqiya Asad



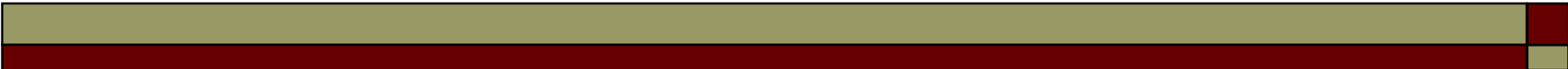
Introduction

Software is the set of instructions in the form of programs to govern the computer system and process the hardware components. To produce a software product a set of activities is used. This set is called a software process.



Personal Software Process (PSP)

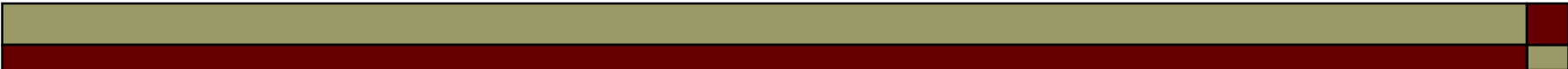
- Focused on **individuals** to improve their software development performance
- Considered an **individual process** rather than a team or organizational process
- Follows a **bottom-up approach** to software process improvement
- **Prescriptive process:** provides clear, structured guidelines
- A **mature methodology** with well-defined **tools and techniques**



Key Features of Personal Software Process (PSP)

Key Features of the Personal Software Process (PSP):

- Process-Oriented:** PSP emphasizes following a structured and disciplined approach throughout software development.
- Individualized:** The process is tailored to each person's skills, experience, and work style, recognizing that everyone has unique strengths and weaknesses.
- Data-Driven:** PSP relies on collecting and analyzing metrics to track progress and identify areas for improvement.
- Stepwise / Incremental:** The development work is divided into smaller, manageable steps, allowing progress to be achieved gradually.
- Quality-Centered:** PSP prioritizes producing high-quality software that meets requirements and minimizes defects.

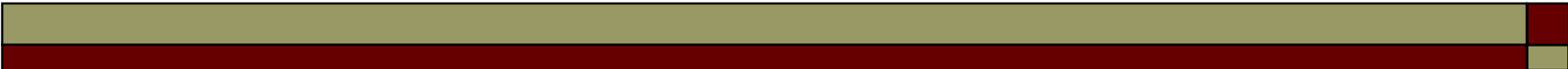


Advantages of Personal Software Process (PSP)

- **Enhanced Productivity:** PSP's structured approach breaks the development process into smaller, manageable steps, helping individuals work more efficiently.
- **Higher Quality:** By tracking and analyzing metrics, PSP helps identify errors early, ensuring software meets requirements and is defect-free.
- **Individualized Approach:** PSP is tailored to a person's skills, experience, and work habits, enabling more effective and efficient work.
- **Better Estimation:** PSP emphasizes accurate planning and estimation, allowing projects to be executed more predictably.
- **Continuous Improvement:** Encourages learning from past projects, fostering ongoing skill development and process enhancement.

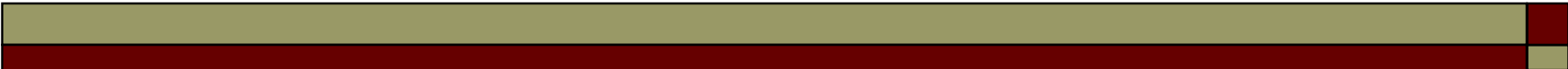
Disadvantages of the Personal Software Process (PSP):

- **Time-Intensive:** PSP can take a lot of time, especially for beginners who need to collect and analyze data throughout the development process.
- **Complexity:** The methodology can be difficult to grasp for individuals with limited experience or knowledge of software engineering principles.
- **Extensive Documentation:** PSP requires thorough documentation at every stage, which can be burdensome and slow down work.
- **Individual Focused:** PSP is primarily designed for personal use, making it less suitable for team-based or collaborative software development projects.



Team Software Process (TSP)

- **Team-based Process:** TSP is designed for teams, unlike PSP which is individual-focused.
- **Focus on Productivity:** Aims to improve overall team productivity and efficiency.
- **Top-Down Approach:** Follows a structured, top-down methodology for process management.
- **Adaptive Process:** TSP can adjust to project needs and team dynamics.
- **Process Management Methodology:** Provides a framework for planning, tracking, and managing team processes.



Key Features of the Team Software Process (TSP):

Team-Oriented: TSP emphasizes collaboration and communication among all team members throughout the software development process.

Process-Driven: Provides a structured and disciplined approach to software development that guides the team step by step.

Data-Driven / Metrics-Focused: Relies on collecting and analyzing metrics to track progress, identify improvement areas, and make informed decisions.

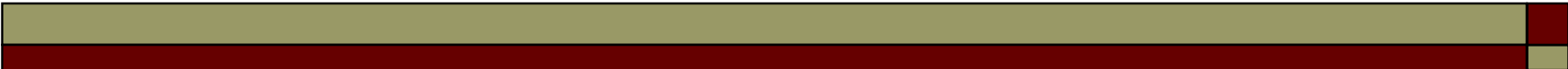
Incremental Approach: Breaks the development process into smaller, manageable steps that can be completed progressively.

Quality-Centered: Focuses on producing high-quality software that meets user requirements and minimizes defects.

Feedback-Oriented: Encourages continuous feedback from peers, mentors, and stakeholders to improve performance and outcomes.

Advantages of the Team Software Process (TSP):

- **Enhanced productivity:** TSP offers a structured framework for software development, helping teams increase productivity by dividing the process into smaller, more manageable tasks.
- **Better quality:** TSP focuses on delivering high-quality software that fulfills user requirements and is largely defect-free. By gathering and analyzing data throughout the development cycle, teams can detect and eliminate errors, improving overall software quality.
- **Stronger team collaboration:** TSP encourages effective teamwork, allowing members to utilize their collective skills and expertise for more efficient project execution.
- **Improved project estimation:** With TSP, accurate estimations are prioritized, enabling teams to plan and carry out projects more reliably.
- **Continuous learning and improvement:** TSP fosters a culture of ongoing improvement, helping teams learn from past experiences and apply those insights to future projects.



Disadvantages of the Team Software Process (TSP):

- **Time-intensive:** TSP can take a lot of time, especially for teams that are new to the methodology and need to gather and analyze data throughout development.
- **Complexity:** TSP can be complicated, particularly for teams with limited software engineering knowledge or less experience in software development.
- **Extensive documentation:** The process involves a large amount of documentation, which can be burdensome for some teams.
- **Requires strict discipline:** TSP demands a disciplined approach to development, which may be difficult for teams that prefer a more flexible workflow.
- **High cost:** Implementing TSP can be expensive, especially if training or software tools are required to support the methodology.

Difference between Personal Software Process (PSP) and Team Software Process (TSP)

Parameters	PSP	TSP
Definition	PSP is a project management process that defines how to manage a project in a face-to-face environment.	TSP is a project management process that defines how to manage a project in a virtual environment.
Formality and Structure	PSP is more formal and structured than TSP.	TSP is less formal and structured than PSP.
Development Model	PSP is based on the waterfall model.	TSP is based on the agile model.
Project Suitability	PSP is more suited for large projects.	TSP is more suited for small projects.
Project Phases	PSP projects are typically completed in one phase.	TSP projects are typically completed in multiple phases.
Language Level	PSP is a high-level language and it is easy to learn and use.	TSP is a low-level language and it is difficult to learn and use.

Difference between Personal Software Process (PSP) and Team Software Process (TSP)

Parameters	PSP	TSP
Language Structure	PSP is a structured language and it is easy to read and write.	TSP is an unstructured language and it is difficult to read and write.
Program Language	PSP programs are written in English and they are easy to understand.	TSP programs are written in assembly language and they are difficult to understand.
Portability	PSP is a portable language and it can be run on any platform.	TSP is a platform-dependent language and it can be run only on specific platforms.
Compilation	PSP is an interpreted language and it does not need to be compiled.	TSP is a compiled language and it needs to be compiled before it can be run.

Difference between Personal Software Process (PSP) and Team Software Process (TSP)

Parameters	PSP	TSP
Cost	PSP is a free language and it can be downloaded from the internet.	TSP is a commercial language and it is not available for free.
Source Availability	PSP is an open-source language and it is available to everyone.	TSP is a closed-source language and it is not available to everyone.
Process Model	PSP is a linear process model that is completed in a sequential manner.	TSP, on the other hand, is an iterative and incremental process model that allows for feedback and changes at each stage of the process.